# Pets4Life Louisville

Assignment 3:
Analysis Modeling
Version 1.0

# Table of Contents


1 Check Volunteer Status #15 ..... 4
1.1 Use Case Descriptions: ..... 4
1.2 System Requirements and Trace Matrix ..... 5
1.2.a System Requirements ..... 5
1.2.b Trace Matrix ..... 5
1.3 Use Case Diagram ..... 6
1.4 Use Case Diagram Narrative ..... 6
1.5 Sequence Diagram ..... 7
1.6 Prototypes ..... 8
2 Contact us (#16-17) ..... 8
2.1 Use case descriptions ..... 8
2.1.a Use Case 16: 'Contact Us' Information ..... 8
2.1.b Use Case 17: 'Contact Us' Request Creation ..... 9
2.2 System Requirements and Trace Matrix ..... 11
2.2.a System Requirements ..... 11
2.2.b Trace Matrix ..... 11
2.3 Use Case Diagram ..... 12
2.4 Use Case Diagram Narrative ..... 12
2.4.a Use Case Diagram Narrative – Use Case 16 ..... 12
2.4.b Use Case Diagram Narrative – Use Case 17 ..... 12
2.5 Sequence Diagram ..... 13
2.5.a Sequence Diagram – Use Case 16 ..... 13
2.5.b Sequence Diagram – Use Case 17 ..... 14
2.6 Prototypes ..... 15
3 Blog Tab #18 ..... 15
3.2 System Requirements and Trace Matrix ..... 16
3.2.a System Requirements ..... 16
3.2.b Trace Matrix ..... 16
3.3 Use Case Diagram ..... 17
3.4 Use Case Diagram Narrative ..... 17
3.5 Sequence Diagram ..... 18
3.6 Prototypes ..... 19
4 Event Customization #19 ..... 19

4.2 System Requirements and Trace Matrix ..... 21
4.2.a System Requirements ..... 21
4.2.b Trace Matrix ..... 21
4.3 Use Case Diagram ..... 21
4.4 Use Case Diagram Narrative ..... 21
4.5 Sequence Diagram ..... 22
4.6 Prototypes ..... 23
5 Volunteering Dashboard #20 ..... 23
5.2 System Requirements and Trace Matrix ..... 24
5.2.a System Requirements ..... 24
5.2.b Trace Matrix ..... 25
5.3 Use Case Diagram ..... 25
5.4 Use Case Diagram Narrative ..... 25
5.5 Sequence Diagram ..... 26
5.6 Prototypes ..... 27
6 Blog Post Customization #21 ..... 27
6.2 System Requirements and Trace Matrix ..... 29
6.2.a System Requirements ..... 29
6.2.b Trace Matrix ..... 29
6.3 Use Case Diagram ..... 29
6.4 Use Case Diagram Narrative ..... 29
6.5 Sequence Diagram ..... 30
6.6 Prototypes ..... 31
7 Class Diagram and Narrative ..... 31
7.1 Class Diagram ..... 31
7.2 Class Diagram Narrative ..... 32

# 1 Check Volunteer Status #15

## 1.1 Use Case Descriptions:

| | | |
|---|---|---|
| **Use Case Name** Check Volunteer Status | **ID**: 15 | **Importance Level**: High |
| **Primary Actor**: User | | **Use Case Type**: Detail, Essential |
| **Stakeholders and Interests**:<br><br>– **User:** Wants easy access to current volunteer event details and their status.<br>– **Administrator:** Wants to ensure event information is accurate and updated to prevent confusion. | | |
| **Brief Description**: Users will be able to check information regarding their volunteering event. | | |
| **Trigger**: User accesses volunteer event information through their dashboard.<br><br>**Type**: External | | |
| **Relationships**:<br><br>Association: User<br><br>Include:<br><br>Extend:<br><br>Generalization: | | |
| **Normal Flow of Events:**<br><br>1. User logs into the website.<br>2. User navigates to the volunteer status page.<br>3. System retrieves and displays the user's volunteering event details and status. | | |
| **SubFlows:** | | |

**Alternate/Exceptional Flows:**

3a. If no volunteering event is found for the user, display a message indicating "No volunteer events found."

## 1.2 System Requirements and Trace Matrix

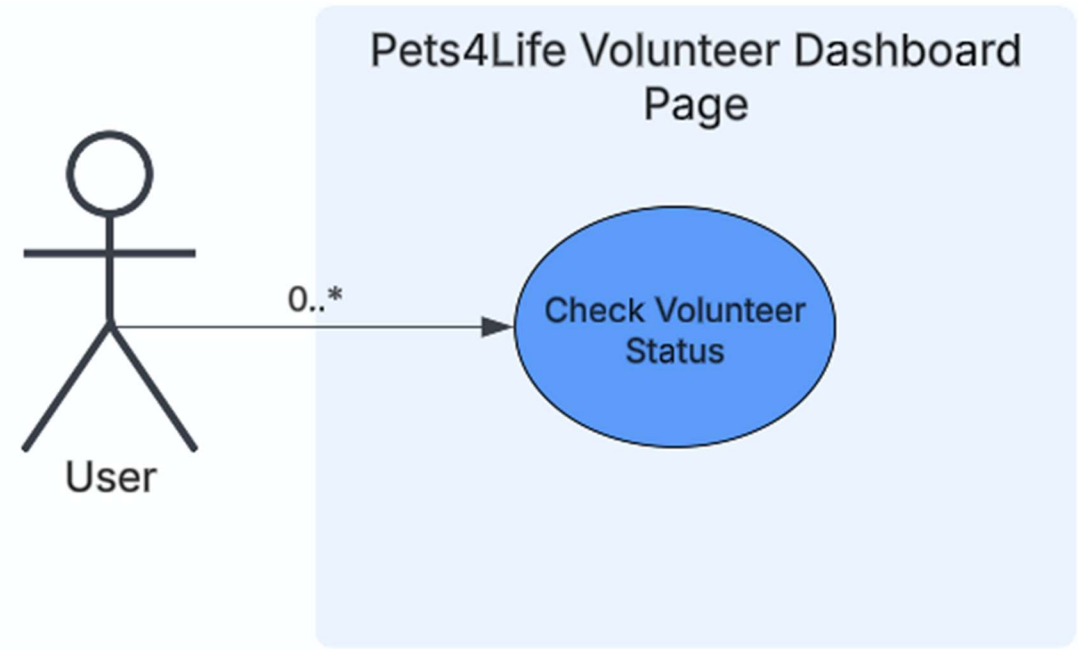
### 1.2.a System Requirements

- ID: SR004 – Volunteer Component will allow users to view and sign up for volunteering events
- ID: SR005 – Volunteer Component will send volunteer email confirmation

### 1.2.b Trace Matrix

| Pets4Life | Check Volunteer Status #15 |
|---|---|
| SR004 – Volunteer Component will allow users to view and sign up for volunteering events | x |
| SR005 – Volunteer Component will send volunteer email confirmation | x |

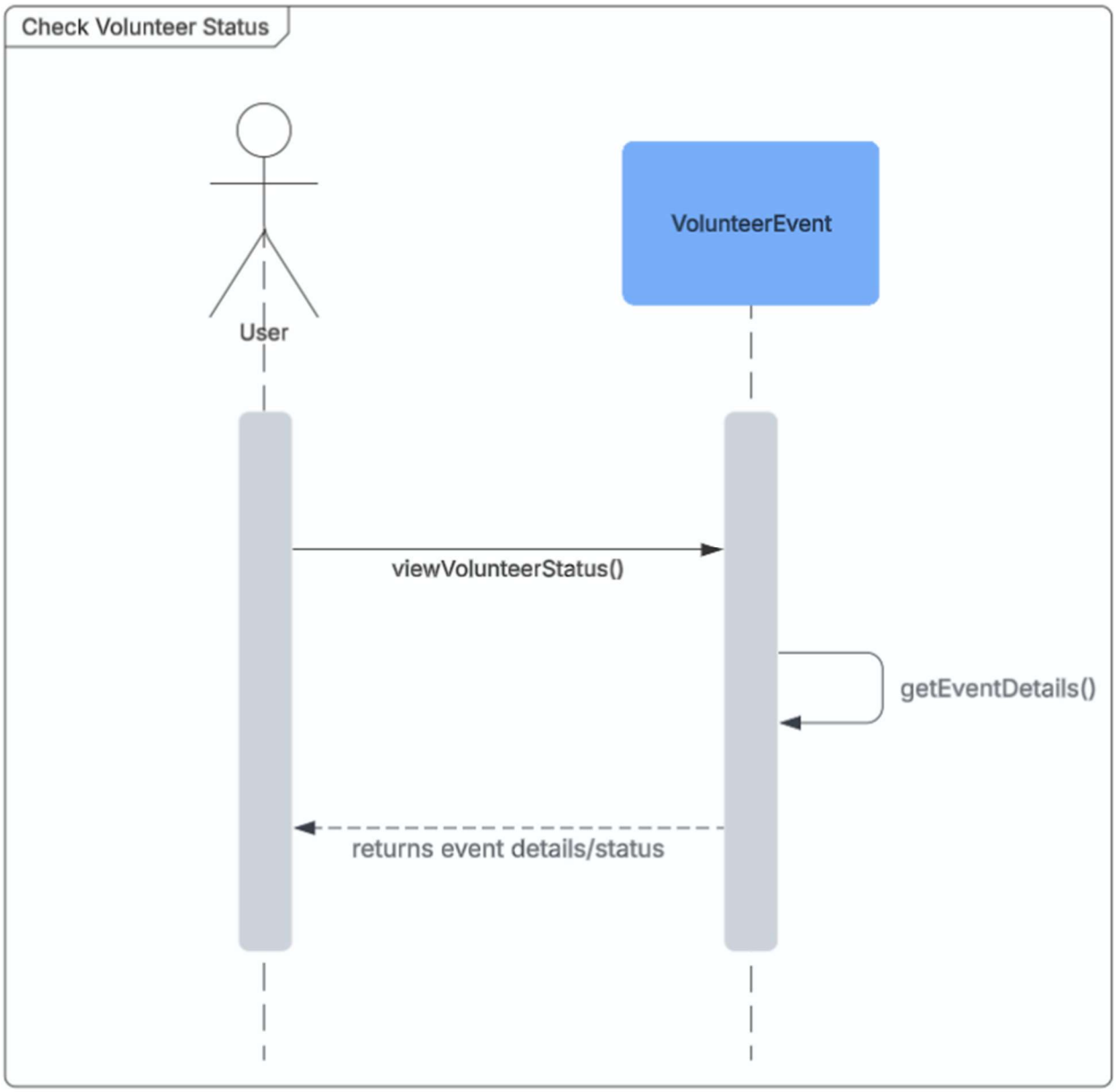
## 1.3 Use Case Diagram



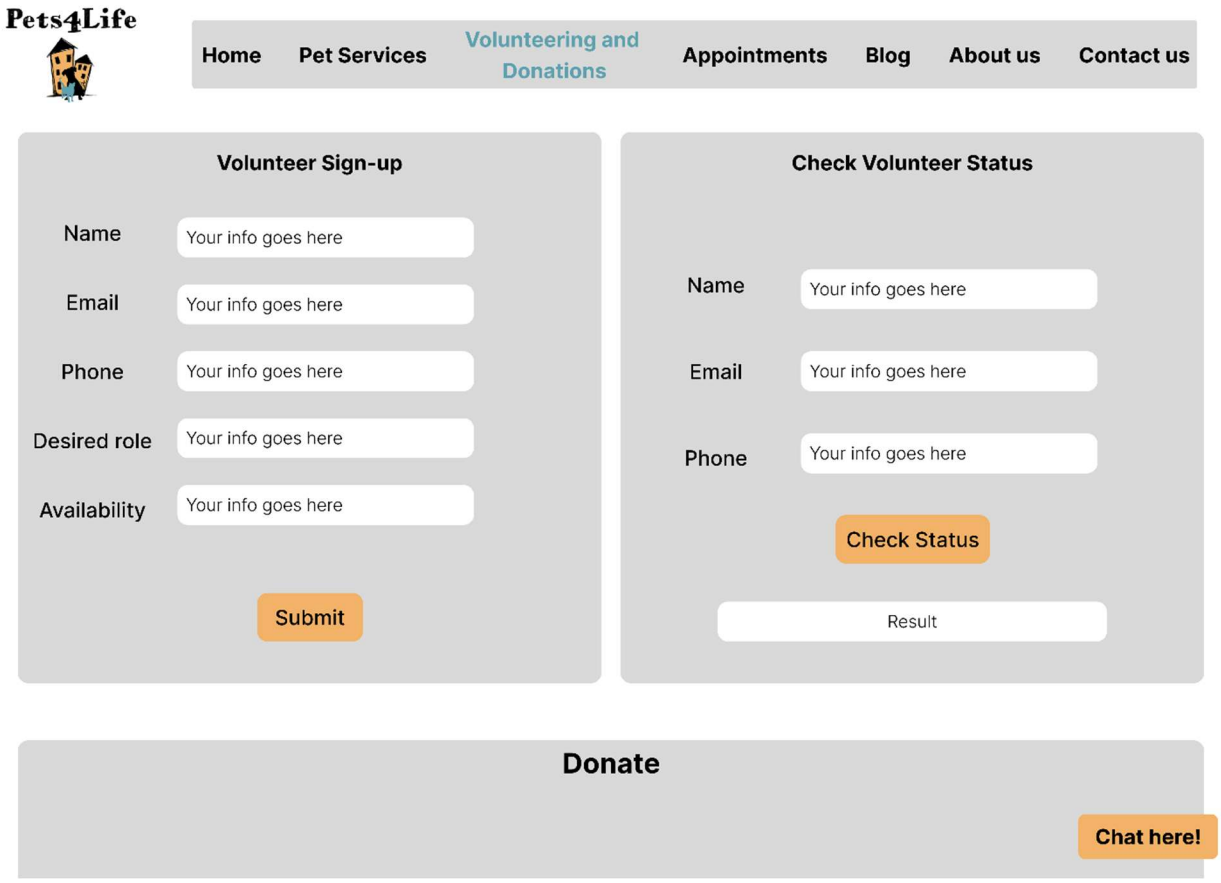

## 1.4 Use Case Diagram Narrative

The use case diagram for Check Volunteer Status shows that the User directly interacts with the Pets4Life Louisville website to access information about volunteering events. This use case ensures transparency and accessibility, allowing volunteers to easily track their commitments and status without administrator intervention.

## 1.5 Sequence Diagram



## 1.6 Prototypes




# 2 Contact us (#16-17)

## 2.1 Use case descriptions

### 2.1.a Use Case 16: 'Contact Us' Information

| | | |
|---|---|---|
| Use Case 16: 'Contact Us' Information | **ID**: 16 | **Importance Level**: High |
| Primary Actor: User | | **Use Case Type**: Detail, Essential |
| **Stakeholders and Interests**:<br><br>– **User:** Needs quick access to essential contact details (phone, email, hours). | | |

| |
|---|
| – **Administrator:** Ensures contact information is accurate, clear, and up-to-date. |
| **Brief Description**: Users can view relevant contact information regarding the organization, such as email, phone, and hours of operation. |
| **Trigger**: User selects the "Contact Us" tab or link on the website.<br>**Type**: External |
| **Relationships**:<br>Association: User<br>Include:<br>Extend:<br>Generalization: |
| **Normal Flow of Events:**<br>1. User navigates to "Contact Us" page/tab.<br>2. System displays organizational contact information (email, phone number, operational hours). |
| **SubFlows:** |
| **Alternate/Exceptional Flows:**<br>2a. If the contact details fail to load, the system displays a default message directing users to check back later or use alternative methods (social media, etc.). |

## 2.1.b Use Case 17: ‘Contact Us’ Request Creation

| | | |
|---|---|---|
| Use Case 17: ‘Contact Us’ Request Creation | **ID**: 17 | **Importance Level**: High |

| Primary Actor: User | **Use Case Type**: Detail, Essential |
|---|---|

**Stakeholders and Interests**:

- **User:** Needs to reliably submit inquiries or requests.
- **Administrator:** Interested in managing and responding to User inquiries efficiently.

**Brief Description**: Users can use the ‘Contact Us’ tab to submit relevant information and create User requests.

**Trigger**: User fills out and submits the "Contact Us" request form.

**Type**: External

**Relationships**:

Association: User

Include:

Extend:

Generalization:

**Normal Flow of Events:**

1. User navigates to the "Contact Us" page.
2. User completes the required fields on the contact form and submits the request.
3. System stores the submitted request.
4. System sends a confirmation message acknowledging receipt.

**SubFlows:**

**Alternate/Exceptional Flows:**

2a. If required fields are incomplete, display an error message requesting completion of required fields.

3a. If storage fails, display an error message advising the user to retry submitting or contact directly by provided phone/email.

## 2.2 System Requirements and Trace Matrix

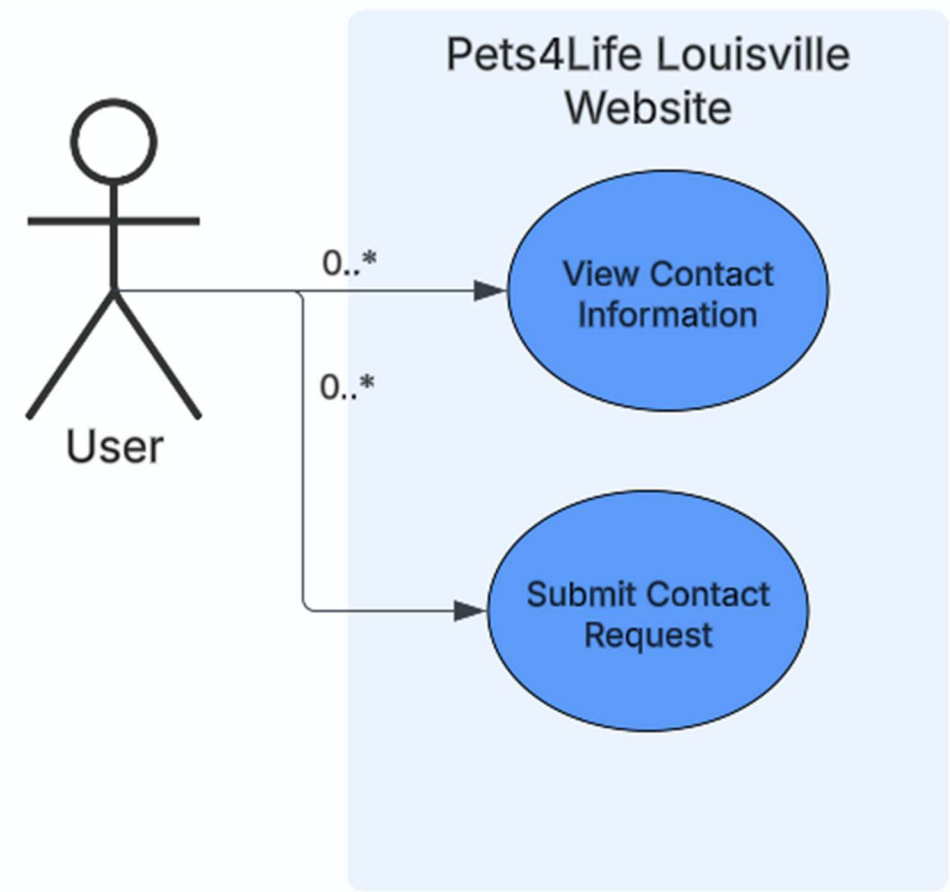
### 2.2.a System Requirements

- ID: SR007 – Contact Us tab will allow User requests to be created
- ID: SR008 – Contact Us tab will display all relevant contact information such as email, phone, and hours of operation

### 2.2.b Trace Matrix

| Pets4Life | 'Contact Us' Information #15 | 'Contact Us' Request Creation #17 |
|---|---|---|
| SR007 – Contact Us tab will allow User requests to be created | x | |
| SR008 – Contact Us tab will display all relevant contact information such as email, phone, and hours of operation | | x |

## 2.3 Use Case Diagram




## 2.4 Use Case Diagram Narrative

### 2.4.a Use Case Diagram Narrative – Use Case 16

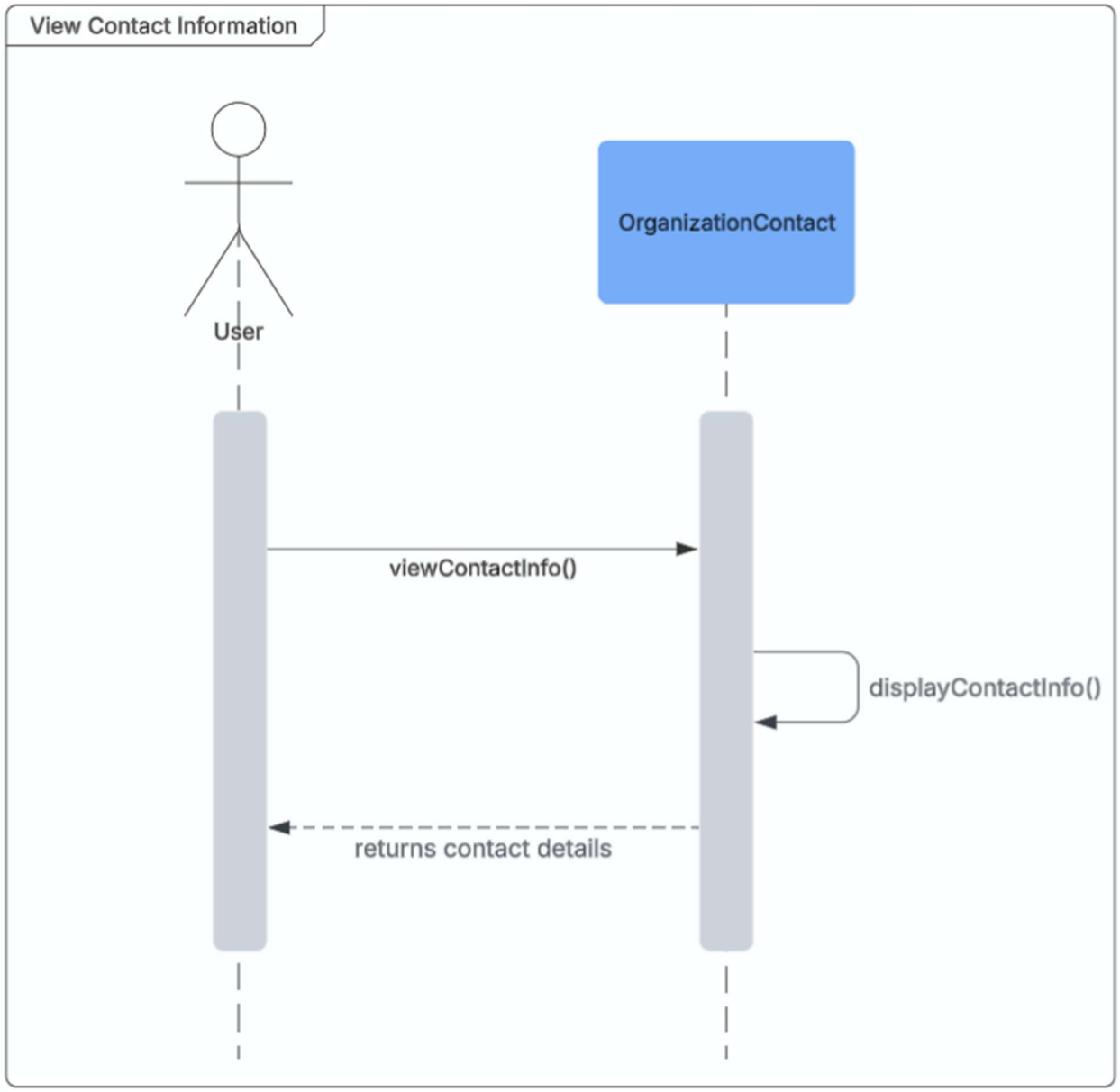
The Contact Us Information use case diagram represents a direct interaction where the User accesses organization details such as phone numbers, emails, and operating hours. It shows the easy user access to important communication channels, creating trust and reliability between Pets4Life Louisville and its community.

### 2.4.b Use Case Diagram Narrative – Use Case 17

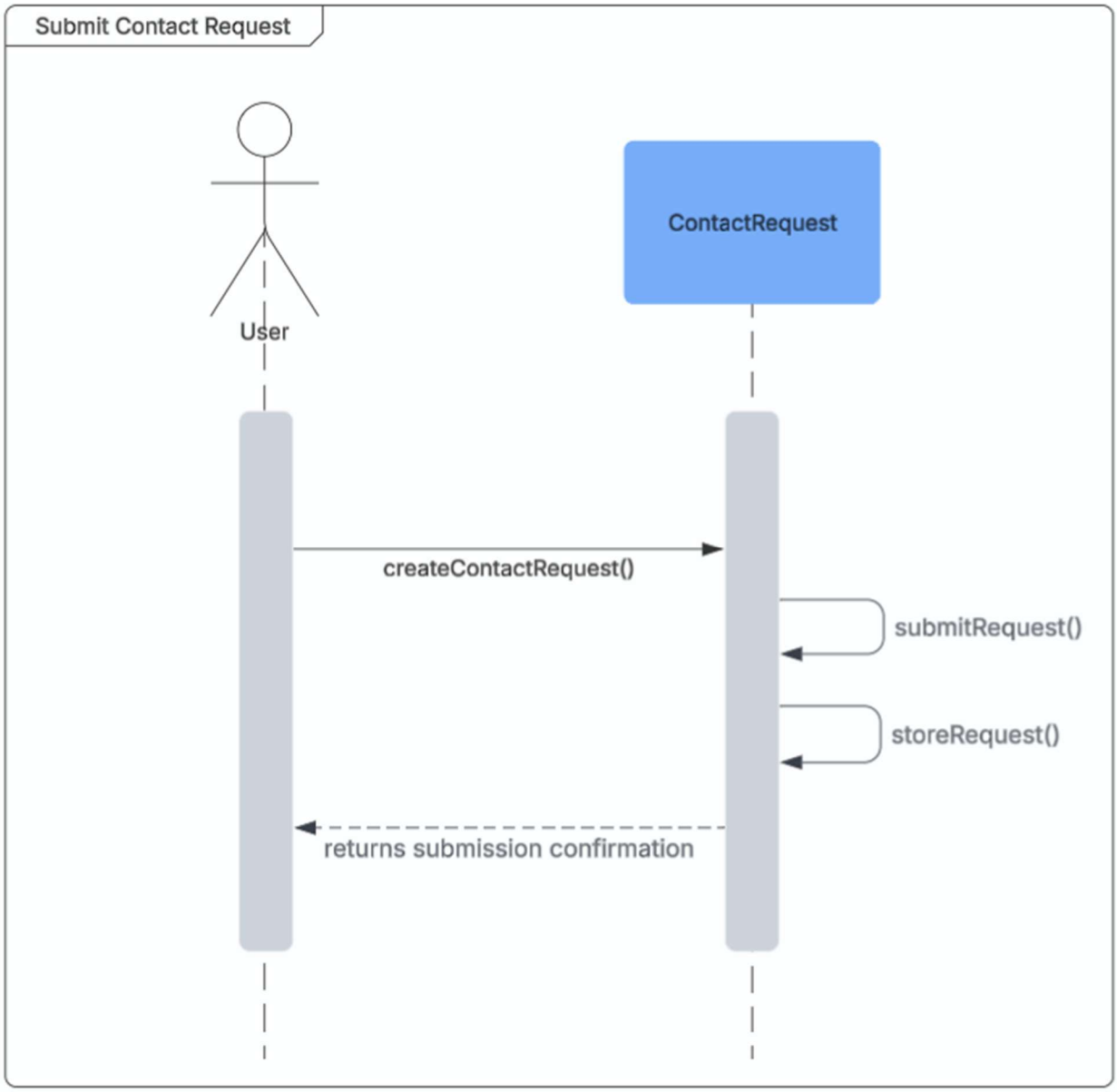
The diagram for Contact Us Request Creation shows clear interaction between the User and the system to submit requests. This ensures efficient communication, streamlining the response process from the organization and improving overall customer support experiences.

## 2.5 Sequence Diagram

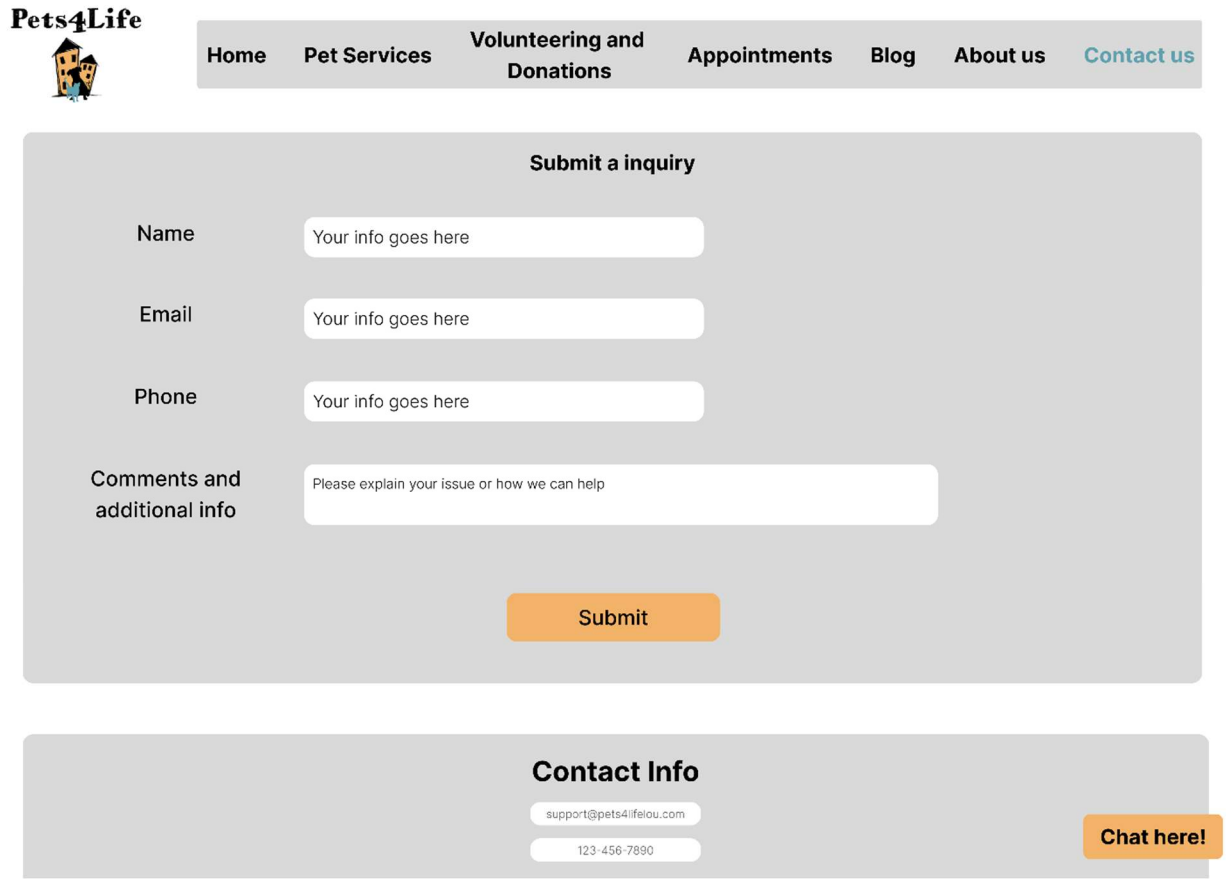
### 2.5.a Sequence Diagram – Use Case 16



## 2.5.b Sequence Diagram – Use Case 17



## 2.6 Prototypes




# 3 Blog Tab #18

| | | |
|---|---|---|
| Use Case 18: Blog Tab | **ID**: 18 | **Importance Level**: High |
| Primary Actor: User | | **Use Case Type**: Detail, Essential |
| **Stakeholders and Interests**:<br><br>– **User:** Wants regular, accessible updates and engaging content from the organization.<br>– **Administrator:** Interested in ensuring content posted effectively communicates relevant organizational news or updates. | | |

| |
|---|
| **Brief Description**: Users can view the 'Blog' tab to find content such as photos & videos, updates and other messages posted from the organization. |
| **Trigger**: User selects the "Blog" tab on the website.<br>**Type**: External |
| **Relationships**:<br>Association: User<br>Include:<br>Extend:<br>Generalization: |
| **Normal Flow of Events:**<br>1. User navigates to the "Blog" tab.<br>2. System retrieves and displays recent blog posts (photos, videos, updates). |
| **SubFlows:** |
| **Alternate/Exceptional Flows:**<br>2a. If no blog posts are available, display a message indicating "No blog posts available." |

## 3.2 System Requirements and Trace Matrix

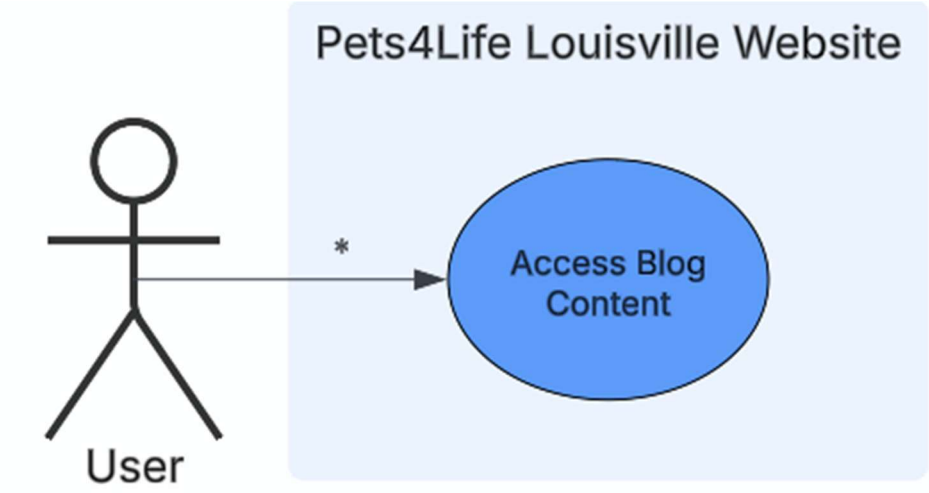
### 3.2.a System Requirements

- ID: SR009 – Blog Tab will display content such as photos, videos, updates and miscellaneous messages from the business

### 3.2.b Trace Matrix

| Pets4Life | Blog Tab #18 |
|---|---|
| SR009 – Blog Tab will display content such as photos, videos, updates and miscellaneous messages from the business | x |

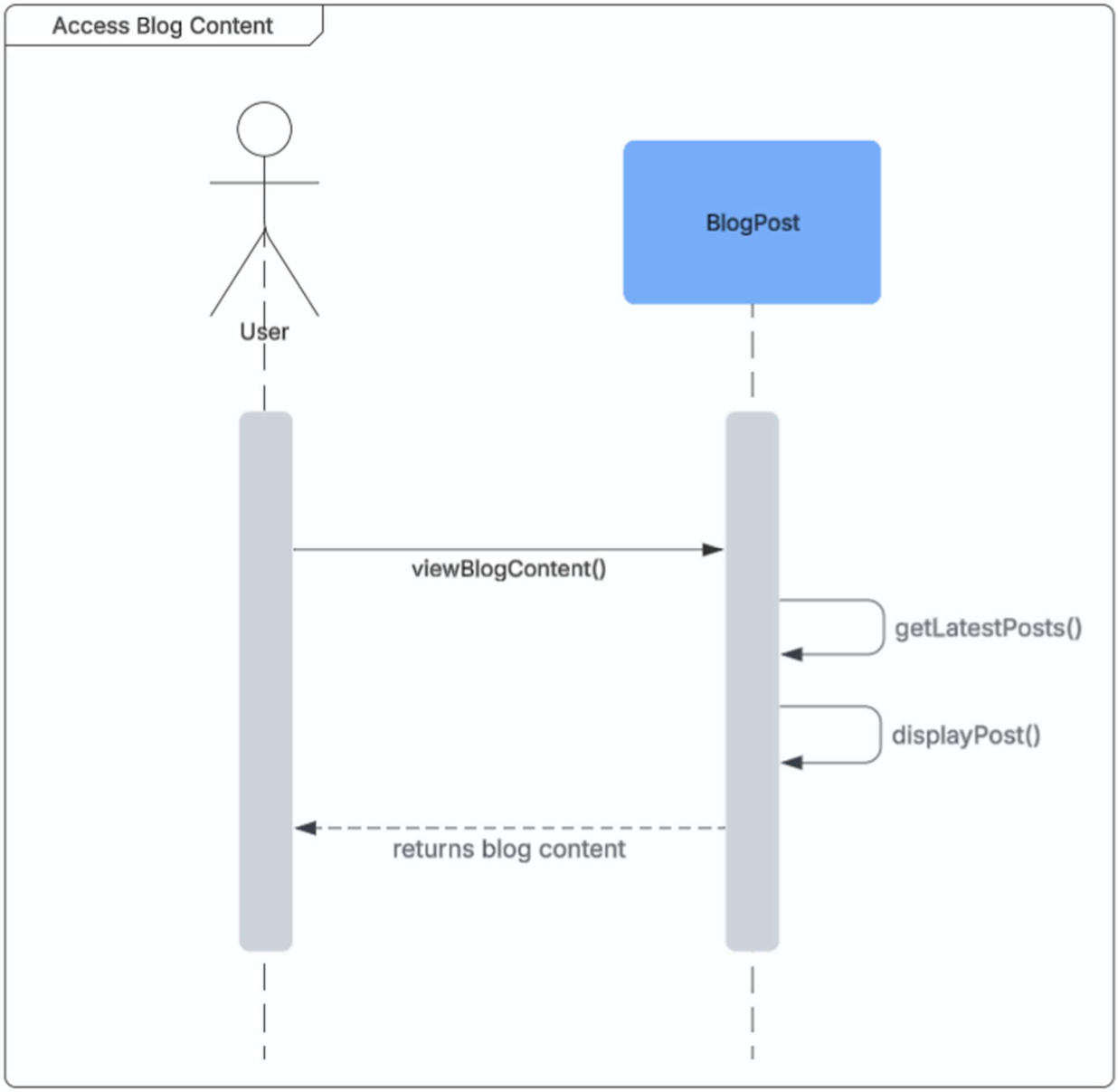
## 3.3 Use Case Diagram



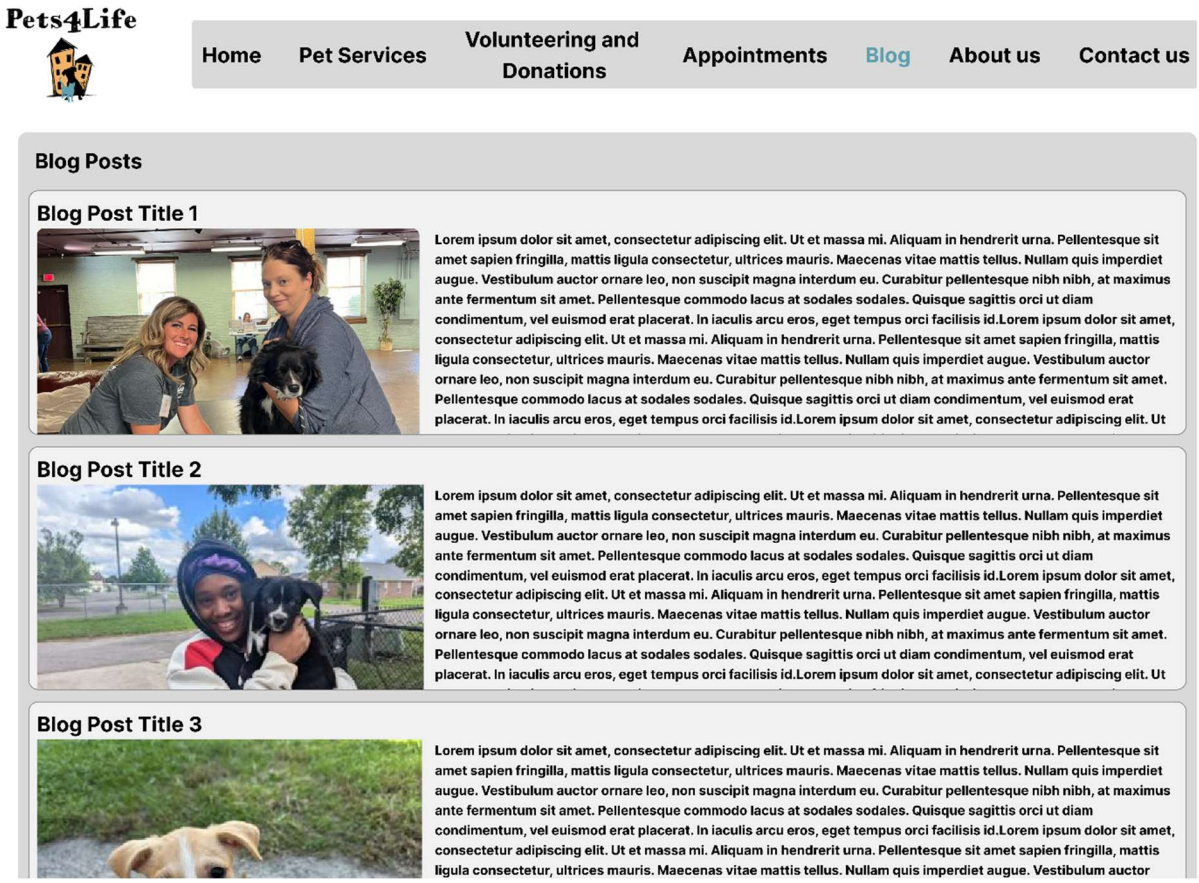

## 3.4 Use Case Diagram Narrative

The Blog Tab use case diagram shows straightforward access by the User to content updates like photos, videos, and announcements. This visual emphasizes the organization's commitment to consistent communication, community engagement, and providing timely updates to its audience.

## 3.5 Sequence Diagram



## 3.6 Prototypes




# 4 Event Customization #19

| | | |
|---|---|---|
| Use Case 19: Event Customization | **ID**: 19 | **Importance Level**: High |
| Primary Actor: Administrator | | **Use Case Type**: Detail, Essential |
| **Stakeholders and Interests**:<br><br>– **User:** Benefits from clear, current, and accurate event information.<br>– **Administrator:** Needs control over event management to organize, create, edit, or delete events effectively. | | |

| |
|---|
| |
| **Brief Description**: Admins can create and host new events, view or edit existing events, and delete or cancel certain events. |
| **Trigger**: Administrator accesses the event management dashboard.<br>**Type**: External |
| **Relationships**:<br>Association: Administrator<br>Include:<br>Extend:<br>Generalization: |
| **Normal Flow of Events:**<br>1. Administrator logs into the admin dashboard.<br>2. Administrator selects event management.<br>3. System displays current events with options to create, edit, or delete events.<br>4. Administrator performs desired action (create/edit/delete).<br>5. System updates events and confirms the update. |
| **SubFlows:**<br>Create: Admin provides new event details and submits.<br>Edit: Admin updates details of existing event and submits.<br>Delete: Admin confirms deletion request. |
| **Alternate/Exceptional Flows:**<br>4a. If submitted event data is invalid or incomplete, system displays an error message prompting corrections. |

| |
|---|
| 5a. If system fails to update due to internal error, display error message advising retry or contact support. |

## 4.2 System Requirements and Trace Matrix

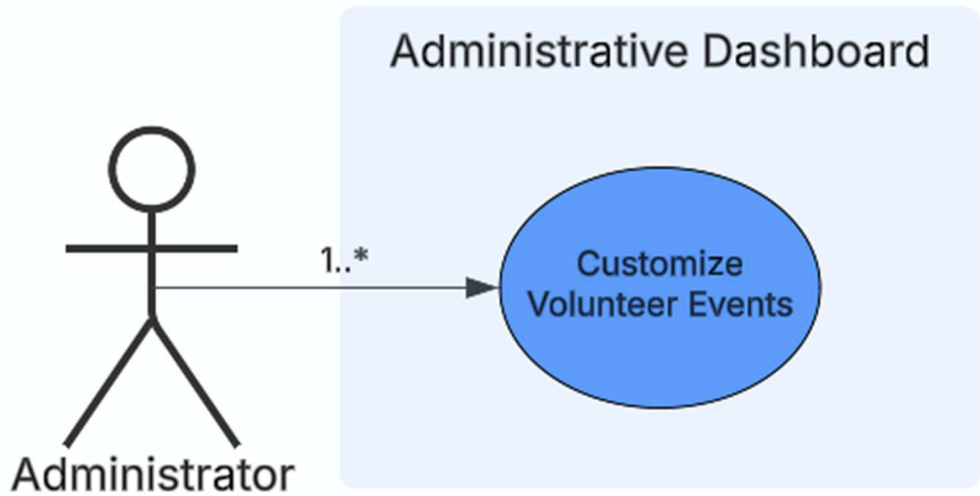
### 4.2.a System Requirements

- ID: SR016 – Employee Dashboard Component will allow the business to add, delete and edit volunteering events

### 4.2.b Trace Matrix

| Pets4Life | Event Customization #19 |
|---|---|
| SR016 – Employee Dashboard Component will allow the business to add, delete and edit volunteering events | x |

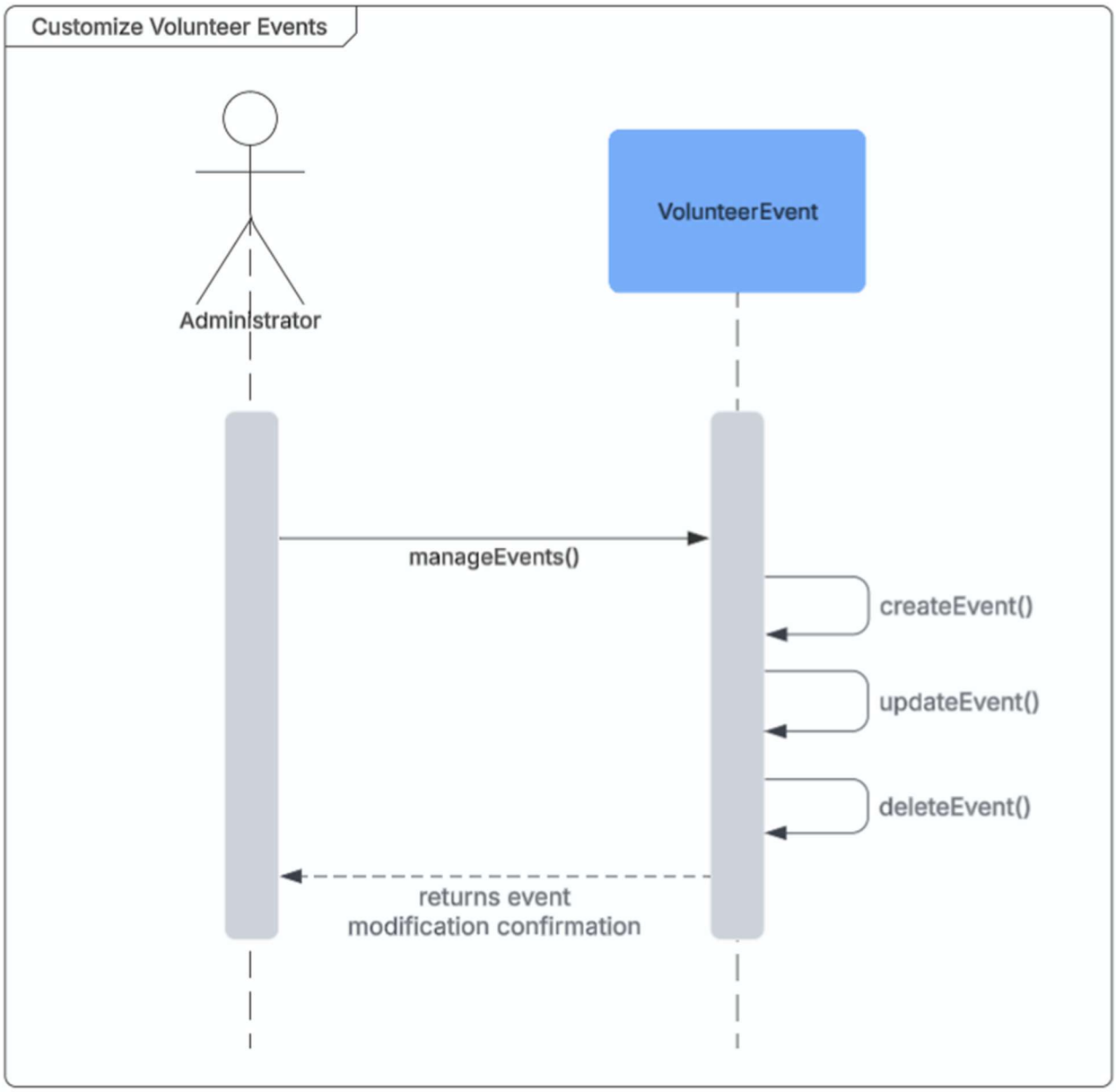
## 4.3 Use Case Diagram



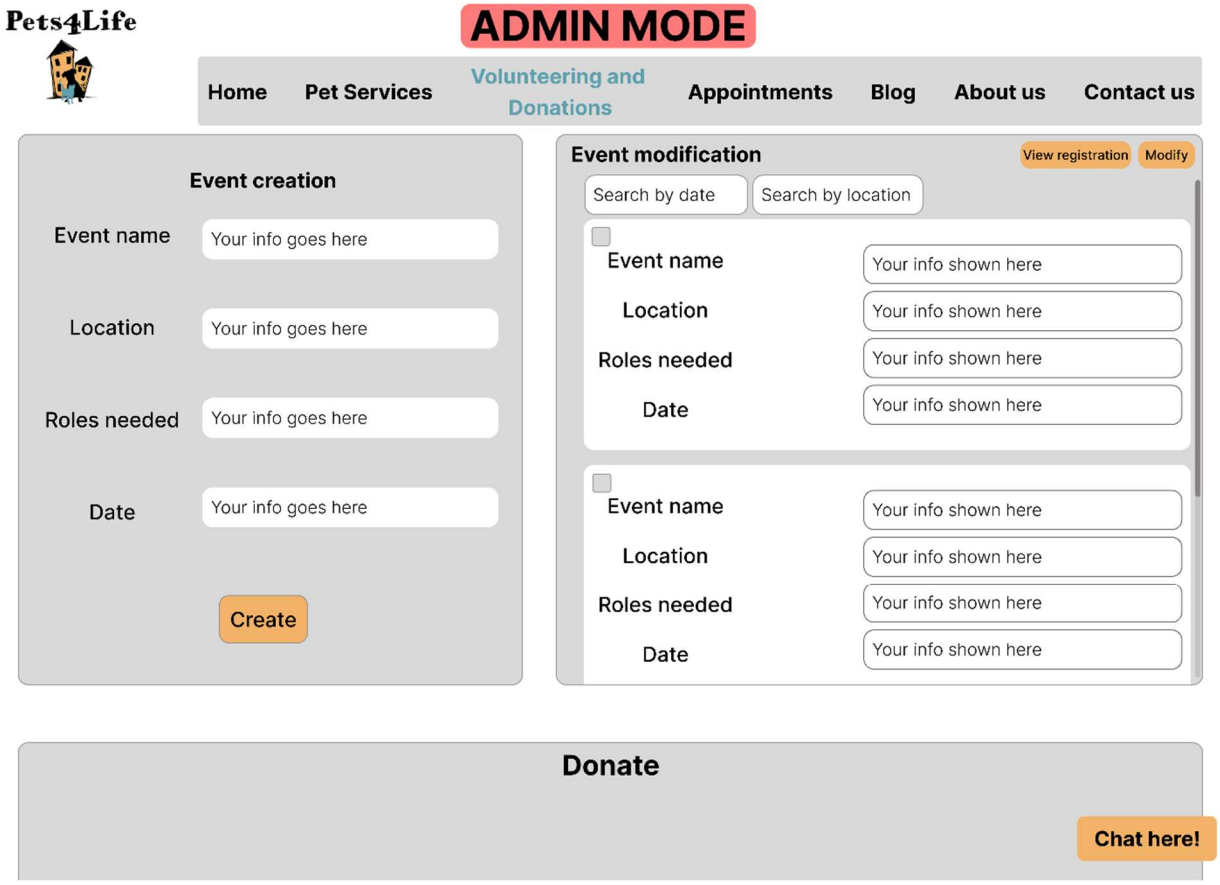

## 4.4 Use Case Diagram Narrative

The Event Customization shows the Administrator interacting directly with the system to create, modify, or delete volunteer events. This ensures the organization maintains accurate, up-to-date event details, allowing for more effective management and smooth operation of volunteering activities.

## 4.5 Sequence Diagram



## 4.6 Prototypes




# 5 Volunteering Dashboard #20

| | | |
|---|---|---|
| Use Case 20: Volunteering Dashboard | **ID**: 20 | **Importance Level**: High |
| Primary Actor: Administrator | | **Use Case Type**: Detail, Essential |
| **Stakeholders and Interests**:<br><br>– **User:** Interested in timely, clear communication about volunteering.<br>– **Administrator:** Requires clear visibility into volunteer participation for effective event management. | | |

| |
|---|
| **Brief Description**: Admins will be able to view all registered events, how many individuals have signed up for an event and will be provided with all of the relevant contact information of each person. |
| **Trigger**: Administrator accesses the volunteering dashboard.<br>**Type**: External |
| **Relationships**:<br>Association: Administrator<br>Include:<br>Extend:<br>Generalization: |
| **Normal Flow of Events:**<br>1. Administrator logs into volunteering dashboard.<br>2. Administrator selects an event to view registrations.<br>3. System retrieves and displays registered volunteer details (names, contact info). |
| **SubFlows:** |
| **Alternate/Exceptional Flows:**<br>3a. If no volunteers are registered, clearly display "No volunteers registered yet." |

## 5.2 System Requirements and Trace Matrix

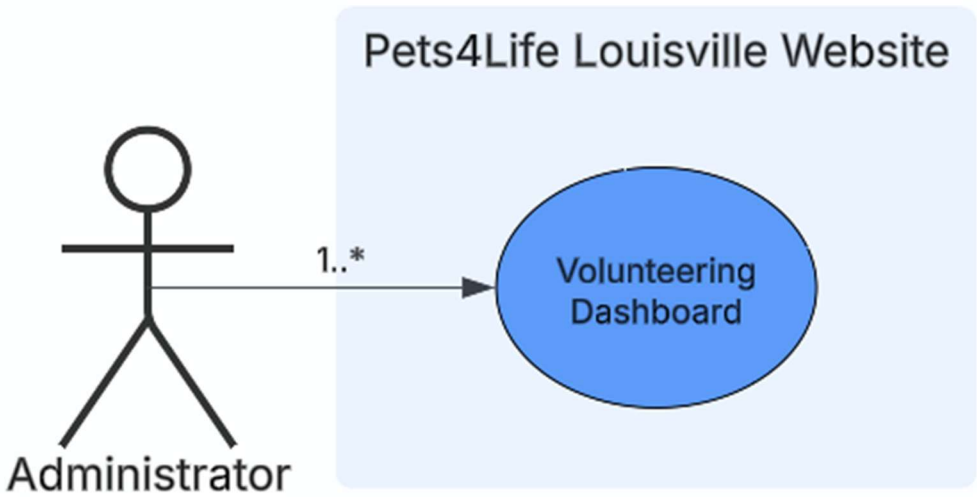
### 5.2.a System Requirements

- ID: SR017 – Employee Dashboard Component will allow the business to view all registered volunteers for events with necessary contact information

## 5.2.b Trace Matrix

| Pets4Life | Volunteering Dashboard #20 |
|---|---|
| SR017 – Employee Dashboard Component will allow the business to view all registered volunteers for events with necessary contact information | x |

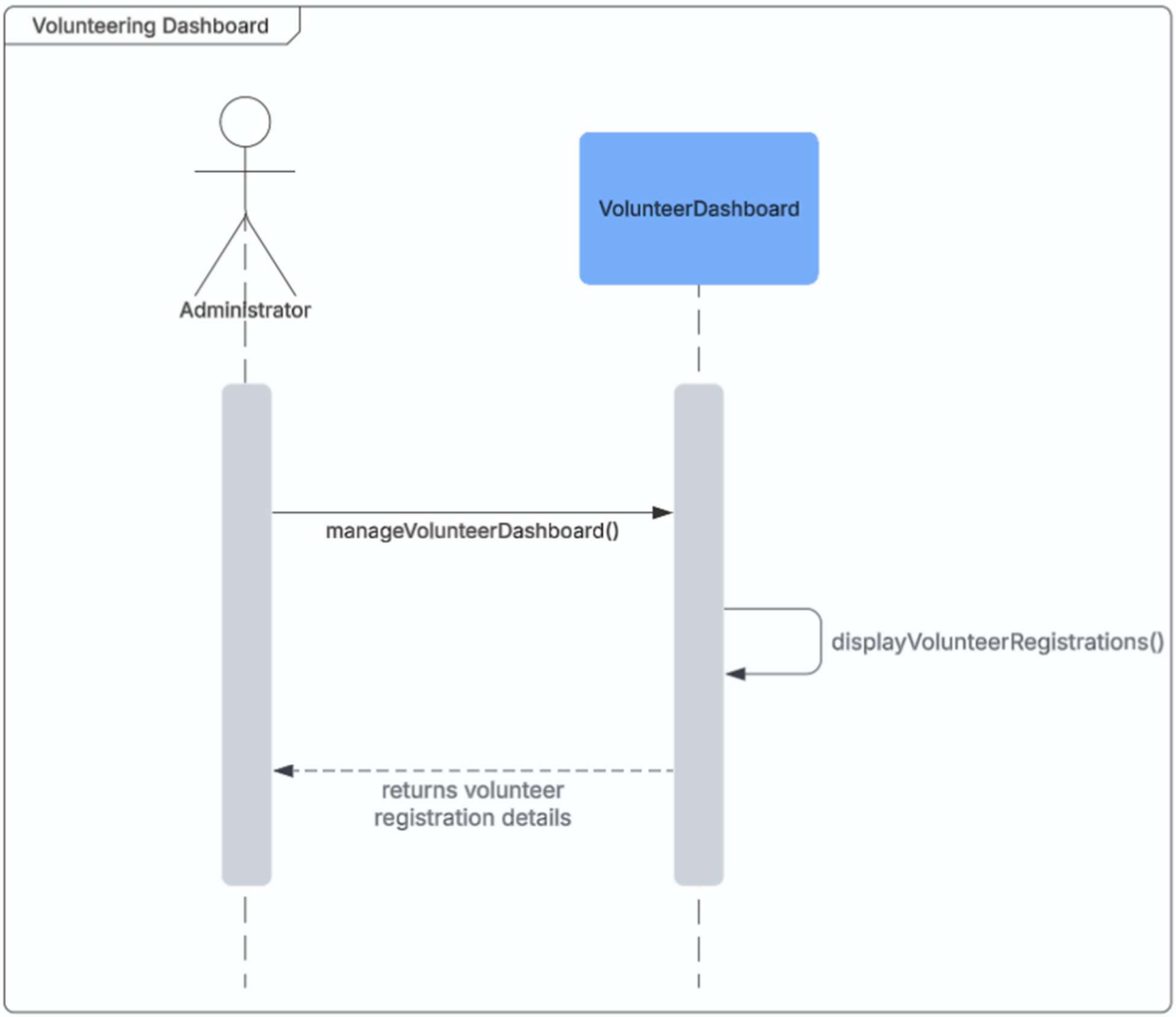
## 5.3 Use Case Diagram



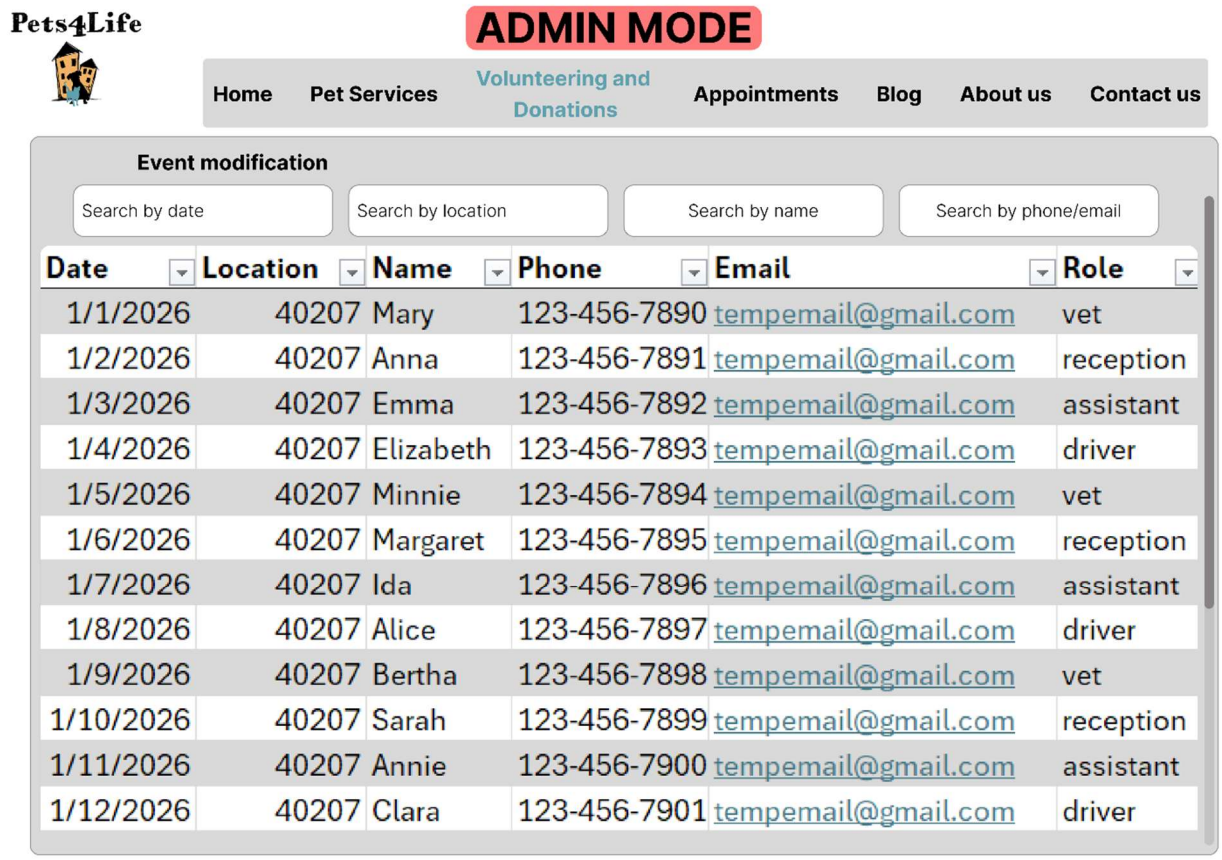

## 5.4 Use Case Diagram Narrative

The use case diagram clearly represents the Administrator's access to the Volunteering Dashboard, enabling viewing and management of volunteer registrations and contact details. This diagram shows the more efficient administrative oversight, simple volunteer coordination and enhanced event organization.

## 5.5 Sequence Diagram



## 5.6 Prototypes




| Date | Location | Name | Phone | Email | Role |
|---|---|---|---|---|---|
| 1/1/2026 | 40207 | Mary | 123-456-7890 | tempemail@gmail.com | vet |
| 1/2/2026 | 40207 | Anna | 123-456-7891 | tempemail@gmail.com | reception |
| 1/3/2026 | 40207 | Emma | 123-456-7892 | tempemail@gmail.com | assistant |
| 1/4/2026 | 40207 | Elizabeth | 123-456-7893 | tempemail@gmail.com | driver |
| 1/5/2026 | 40207 | Minnie | 123-456-7894 | tempemail@gmail.com | vet |
| 1/6/2026 | 40207 | Margaret | 123-456-7895 | tempemail@gmail.com | reception |
| 1/7/2026 | 40207 | Ida | 123-456-7896 | tempemail@gmail.com | assistant |
| 1/8/2026 | 40207 | Alice | 123-456-7897 | tempemail@gmail.com | driver |
| 1/9/2026 | 40207 | Bertha | 123-456-7898 | tempemail@gmail.com | vet |
| 1/10/2026 | 40207 | Sarah | 123-456-7899 | tempemail@gmail.com | reception |
| 1/11/2026 | 40207 | Annie | 123-456-7900 | tempemail@gmail.com | assistant |
| 1/12/2026 | 40207 | Clara | 123-456-7901 | tempemail@gmail.com | driver |

# 6 Blog Post Customization #21

| | | |
|---|---|---|
| Use Case 21: Blog Post Customization | **ID**: 21 | **Importance Level**: High |
| Primary Actor: Administrator | | **Use Case Type**: Detail, Essential |
| **Stakeholders and Interests**:<br>– **User:** Expects current, engaging, and relevant blog content.<br>– **Administrator:** Needs capability to regularly update, manage, and maintain blog content easily. | | |

**Brief Description**: Admins will have the ability to add new blog posts, edit existing ones or delete certain posts.

**Trigger**: Administrator selects blog management on dashboard.

**Type**: External

**Relationships**:

Association: Administrator

Include:

Extend:

Generalization:

**Normal Flow of Events:**

1. Administrator logs into admin dashboard.
2. Administrator navigates to blog post management.
3. System shows existing blog posts with options to create, edit, or delete.
4. Administrator chooses desired action (create/edit/delete).
5. System updates the blog content and confirms the update.

**SubFlows:**

Create: Admin inputs details for new blog post and submits.

Edit: Admin modifies existing blog post and submits changes.

Delete: Admin confirms deletion of selected blog post.

**Alternate/Exceptional Flows:**

4a. If blog content details are incomplete or invalid, system returns an error message to administrator to correct the issues.

5a. If an internal error prevents the blog post update, system advises administrator to retry or contact technical support.

## 6.2 System Requirements and Trace Matrix

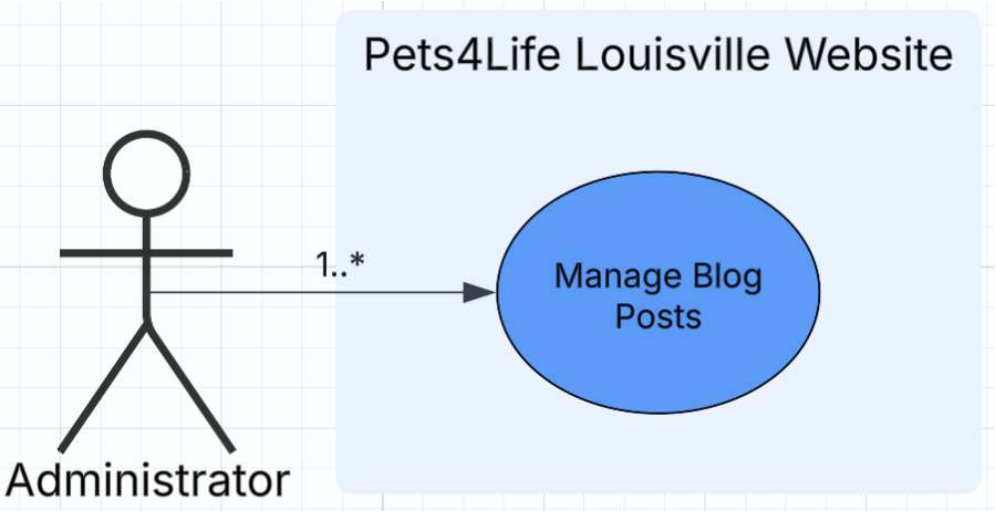
### 6.2.a System Requirements

- ID: SR018 – Employee Dashboard Component will allow the business to add, delete and edit blog posts

### 6.2.b Trace Matrix

| Pets4Life | Blog Post Customization #21 |
|---|---|
| SR018 – Employee Dashboard Component will allow the business to add, delete and edit blog posts | X |

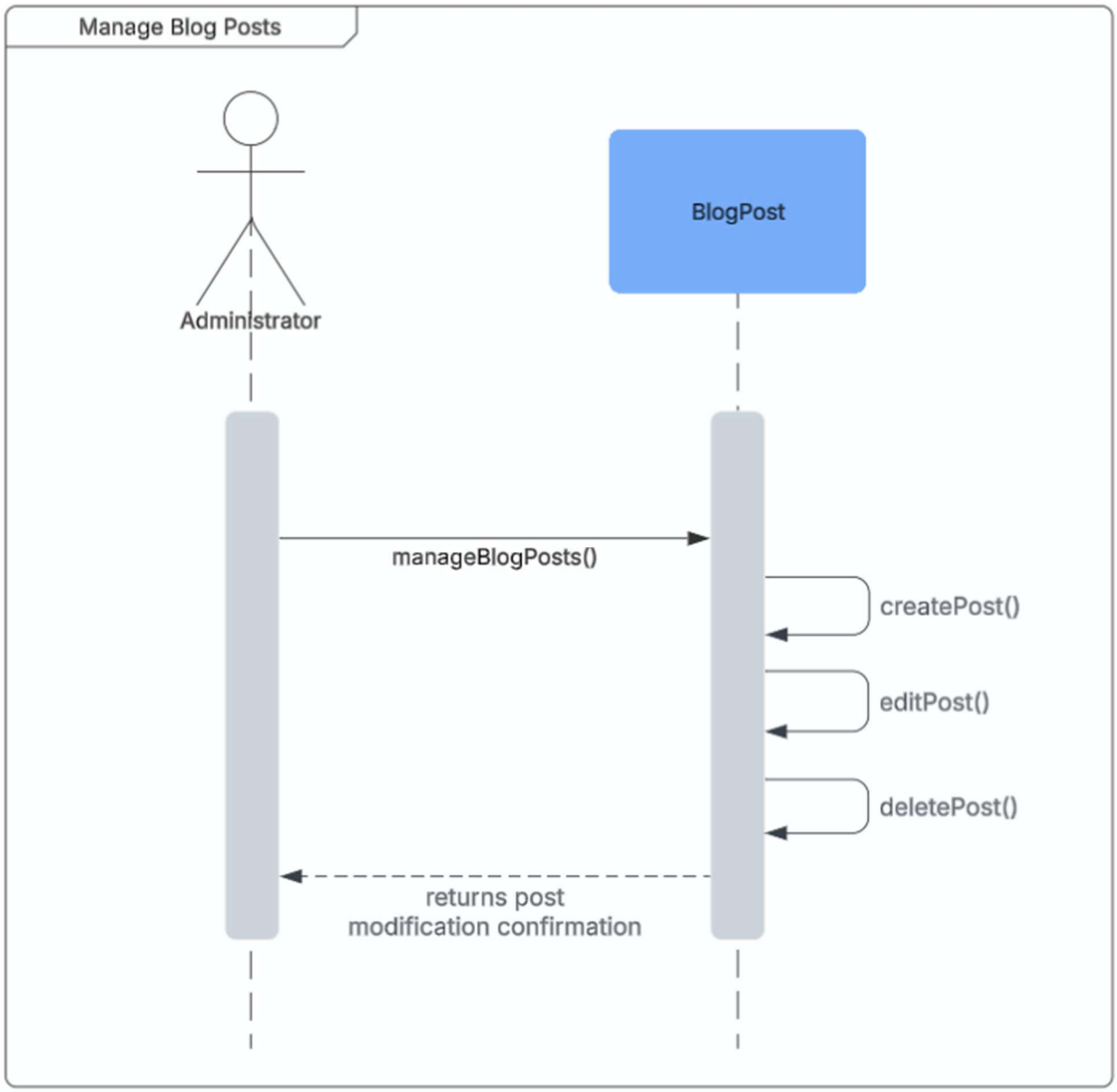
## 6.3 Use Case Diagram



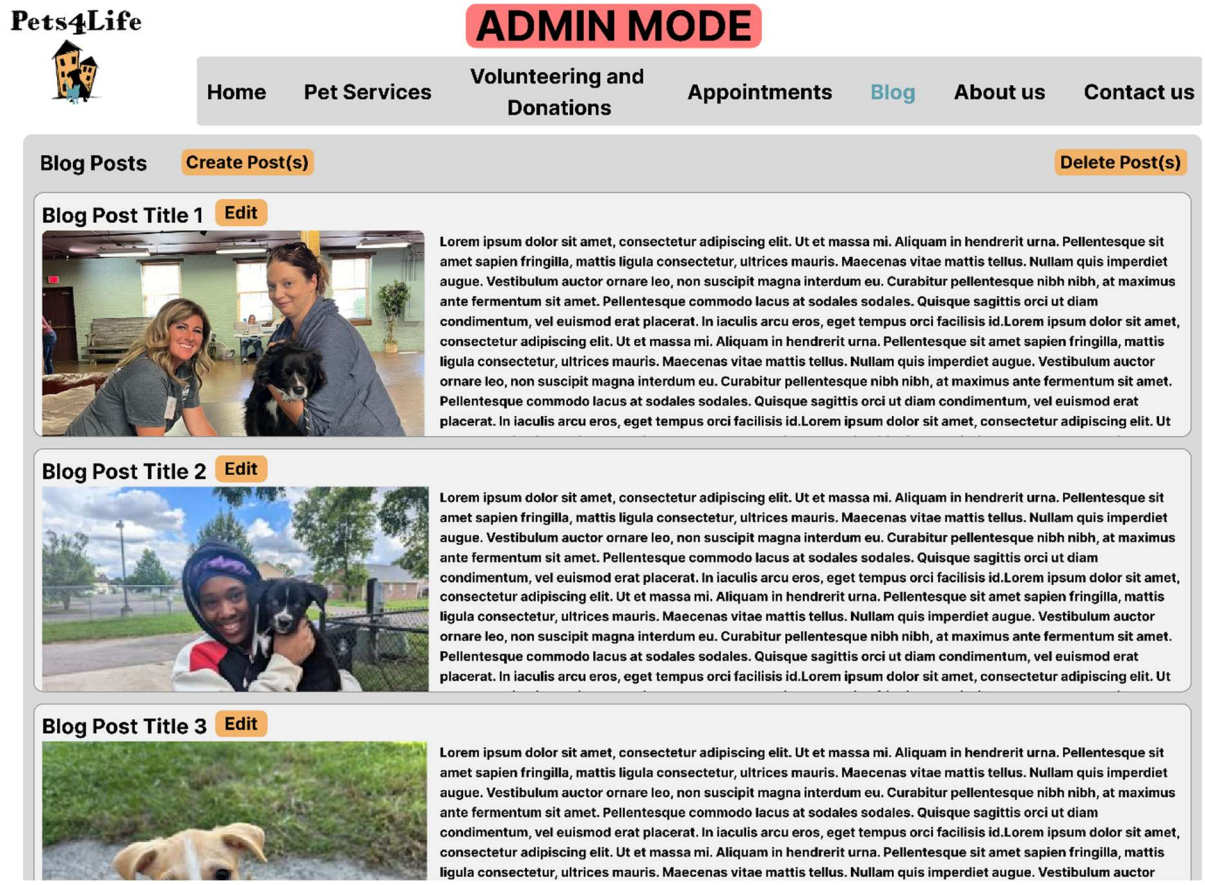

## 6.4 Use Case Diagram Narrative

The diagram for Blog Post Customization shows the direct management actions available to the Administrator, including creating, editing, or deleting blog posts. This reflects the organization's to easily maintain relevant, engaging, and up-to-date content for the community.

## 6.5 Sequence Diagram



## 6.6 Prototypes

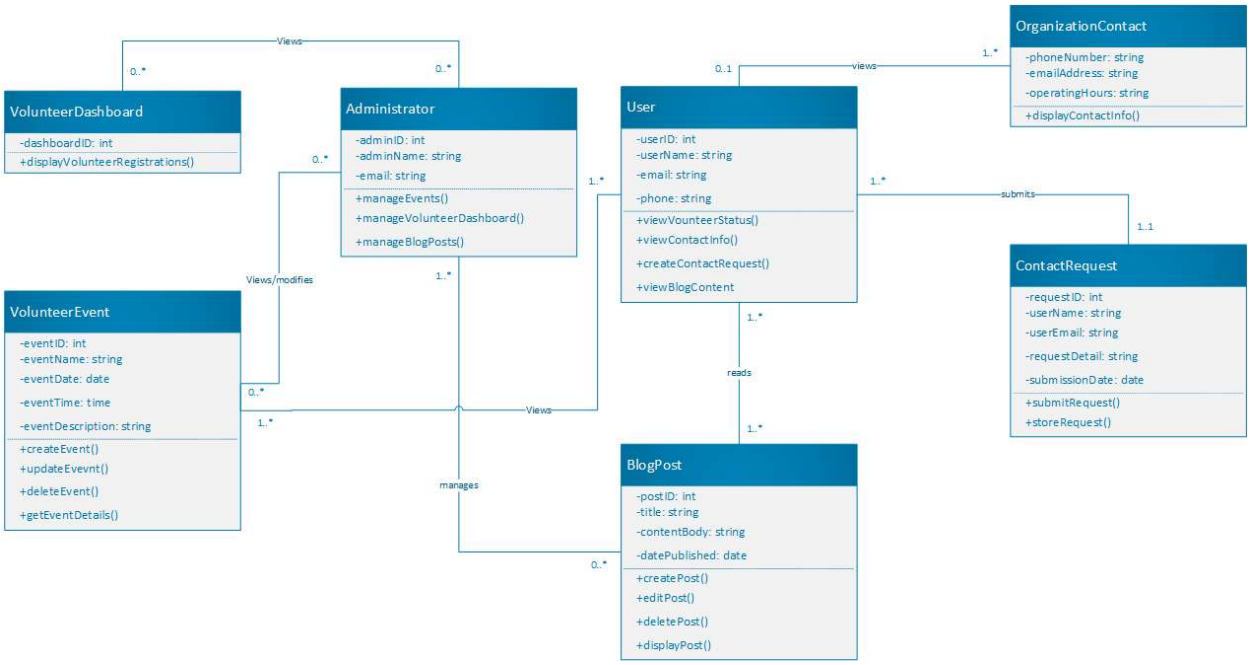



# 7 Class Diagram and Narrative

## 7.1 Class Diagram

**Overall Class Diagram Description (All Assigned Use Cases)**

The class diagram should visually represent the core structure of the Pets4Life Louisville system, clearly identifying key entities (classes), their attributes, and methods relevant to your assigned use cases. Relationships should also clearly reflect the interactions between these classes.




## 7.2 Class Diagram Narrative

Interpretation Narrative:

This class diagram represents the structural overview of the Pets4Life Louisville system. It defines clear roles (User and Administrator), clearly delineates responsibilities, and provides structured ways these roles interact with essential entities. Each class includes key attributes and methods vital to the business operations, clarifying both user interactions (e.g., viewing volunteer status or submitting requests) and administrator management tasks (e.g., event and blog management).

This visual representation will help stakeholders easily identify and understand data handling and functional processes clearly within the system, facilitating efficient communication, system design, and subsequent development phases.